COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S3
DATE PERFORMED:	10/6/26	**DATE SUBMITTED:**	10/6/26
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Seatwork 10.1 **Graph**

1. PROCEDURE OUTPUT

ILO A: Create C++ code for graph implementation utilizing adjacency matrix and adjacency list

Code Output:

```
Graph adjacency list
(start_vertex, end_vertex, weight):
(0, 2, 4) (0, 1, 2)
(1, 4, 3)
(2, 3, 2)
(3, 1, 4)
(4, 3, 3)
```

ILO B: Create C++ code for implementing graph traversal algorithms such as Breadth-First and Depth-First Search

Code Output of A:

```
1:      {2: 0}, {5: 0},
2:      {1: 0}, {5: 0}, {4: 0},
3:      {4: 0}, {7: 0},
4:      {2: 0}, {3: 0}, {5: 0}, {6: 0}, {8: 0},
5:      {1: 0}, {2: 0}, {4: 0}, {8: 0},
6:      {4: 0}, {7: 0}, {8: 0},
7:      {3: 0}, {6: 0},
8:      {4: 0}, {5: 0}, {6: 0},

DFS Order of vertices:
1
5
8
6
7
3
4
2
```

Code Output of B:

```
1:      {2: 2}, {5: 3},
2:      {1: 2}, {5: 5}, {4: 1},
3:      {4: 2}, {7: 3},
4:      {2: 1}, {3: 2}, {5: 2}, {6: 4}, {8: 5},
5:      {1: 3}, {2: 5}, {4: 2}, {8: 3},
6:      {4: 4}, {7: 4}, {8: 1},
7:      {3: 3}, {6: 4},
8:      {4: 5}, {5: 3}, {6: 1},

BFS Order of vertices:
1
2
5
4
8
3
6
7
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

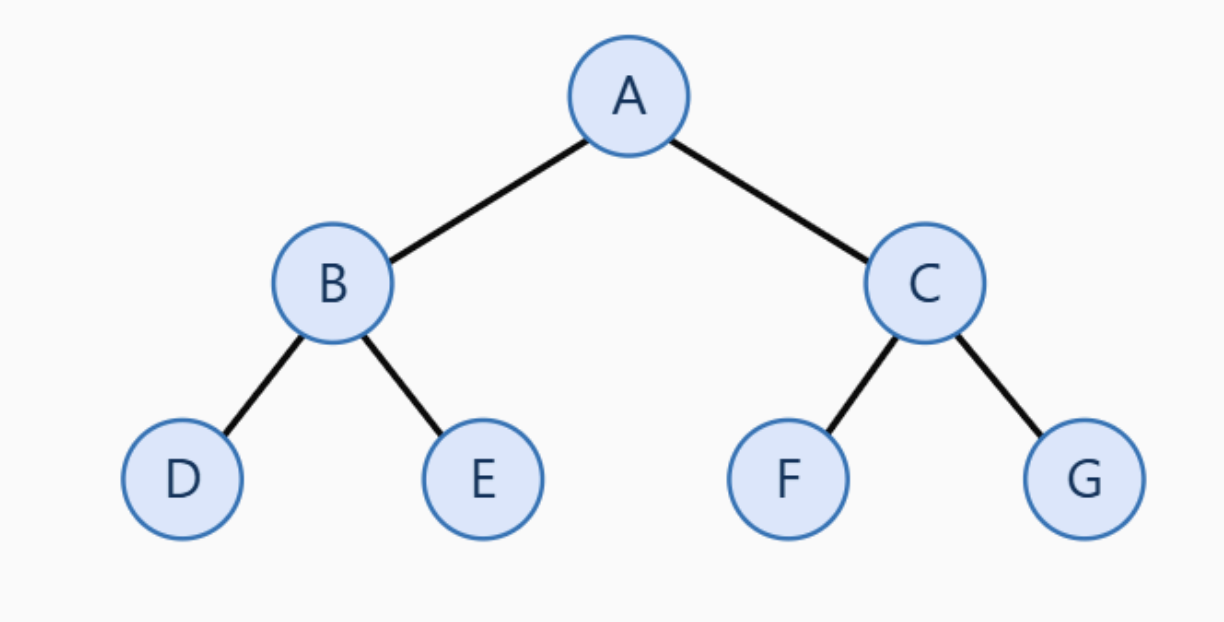
ILO C: Demonstrate an understanding of graph implementation, operations and traversal methods.

1. A person wants to visit different locations indicated on a map. He starts from one location (vertex) and wants to visit every vertex until it finishes from one vertex, backtracks, and then explore other vertex from same vertex. Discuss which algorithm would be most helpful to accomplish this task.

DFS (Depth-First Search) is the best algorithm for this task because it explores one path as far as it can before going back to the previous vertex. After backtracking, it will explore other unvisited vertices until it finishes.

2. Identify the equivalent of DFS in traversal strategies for trees. To efficiently answer this question, provide a graphical comparison, examine pseudocode and code implementation.

The equivalent of DFS in tree traversal is depth-first tree traversal. It has three common methods which are Pre-order, In-order, and Post-order. They all explore the tree deeply first, but the difference is when they visit the root node.



Pre-order: A, B, D, E, C, F, G
In-order: D, B, E, A, F, C, G
Post-order: D, E, B, F, G, C, A

```
DFS(vertex):
    mark vertex as visited
    for each neighbor of vertex:
        if neighbor is not visited:
            DFS(neighbor)
```

3. In the performed code, what data structure is used to implement the Breadth First Search?

The data structure used in the BFS code is a queue. It follows the First In, First Out (FIFO) rule, meaning the vertex that enters first will also be processed first.

4. How many times can a node be visited in the BFS?

In BFS, a node should only be visited once. The visited list or set is used to prevent visiting and processing the same node again.

Conclusion:

Summary of the lesson learned:
In this activity, I learned how graphs work and how vertices and edges are connected. I also learned how BFS visits the vertices using a queue, starting from one vertex and moving to its nearby connected vertices.

Analysis of the procedure:

The procedure helped me understand how to create a graph and show its connections using an adjacency list. At first, the code can be a little confusing but after understanding how the vertices are connected it became easier to follow how BFS works.

Analysis of the supplementary activity:

The supplementary activity helped me understand the difference between BFS and DFS. BFS visits nearby vertices first using a queue while DFS explores one path first before going back. I also learned the different methods of tree traversal and how they visit each node.

Concluding statement / Feedback:

I think I did quite okay in this activity because I was able to understand the basic concept of graphs and BFS. There are still some parts of the code that I need to practice more especially when implementing graph traversal by myself.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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